



FACULTY OF BUSINESS AND IT

MBAI 5600G - Applied Integrative Analytics Capstone Project

Milestones 6

Due: July 21, 2026

Milestone	Description	Due Date
1	Capstone project selection/proposal	May 19
2	Literature review and data source identification	May 26
3	Data collection & Initial EDA	June 9
4	Data Preprocessing & Baseline Modeling	June 23
5	Advanced Modeling & Optimization	July 7
6	Validation & Business Impact Assessment	July 21
7	Capstone project oral presentation	July 27 & August 10
8	Capstone project report	August 10

Important Note on Pace and Deadlines

Project complexity and team progress may vary throughout the term. If your group completes a milestone before its scheduled deadline, you are encouraged to begin working on the next milestone immediately rather than waiting for the due date.

The posted deadlines represent the maximum allowable completion time for each milestone and should not be interpreted as required pacing checkpoints. Early progress is strongly encouraged to allow additional time for refinement, testing, validation, and project improvement.

Milestone 6: Validation & Business Impact Assessment

Milestone 6 focuses on validating the optimized model developed in Milestone 5 and assessing its practical value within the context of the selected business problem. At this stage, students should move beyond reporting predictive performance and demonstrate that the proposed solution is reliable, robust, interpretable, and capable of supporting informed business decision-making.

The objective of this milestone is to determine whether the selected **model generalizes well to unseen data, produces consistent and trustworthy results, and creates measurable business value**. Students should critically evaluate model robustness, identify limitations, assess implementation feasibility, and translate technical findings into actionable business insights and recommendations.

The validation and business impact assessment should clearly explain:

- How the final model was validated,
- Whether the model generalizes well to unseen data,
- How robust the model is under different conditions,
- Whether model predictions are reliable and explainable,
- How the model addresses the original business problem,
- The expected business impact of model deployment,
- Potential implementation challenges and risks,
- Recommendations for practical deployment and future improvements.

The primary objective of Milestone 6 is to validate the final modeling solution, demonstrate its business relevance, and justify whether it is suitable for real-world implementation.

1. Model Validation

Students must perform a comprehensive validation of the final selected model. Students must clearly document:

- Validation methodology used,
- Validation dataset(s),
- Cross-validation or resampling procedures (if applicable),
- Hold-out or test set evaluation,
- Model generalization performance,
- Evidence supporting model reliability.

2. Robustness and Reliability Assessment

Students must evaluate the robustness and stability of the proposed model. Students should investigate:

- Consistency of model performance across different datasets or validation folds,
- Sensitivity to input variations,
- Performance under noisy or incomplete data (where applicable),
- Stability of predictions,
- Model reliability under realistic operating conditions.

3. Error Analysis

Students must perform a detailed analysis of model errors. Students should identify:

- Common prediction errors,
- Misclassified observations (for classification projects),
- Large prediction residuals (for regression projects),
- Patterns associated with incorrect predictions,
- Possible causes of model failure,
- Opportunities for future improvement.

4. **Business Impact Assessment**

Students must evaluate the practical implications of deploying the final model. Students should discuss:

- How the model supports business decision-making,
- Expected operational improvements,
- Financial or organizational benefits,
- Potential cost savings,
- Revenue opportunities,
- Productivity improvements,
- Customer or stakeholder impact,
- Alignment with organizational objectives.

5. **Deployment Feasibility Assessment**

Students must evaluate whether the proposed solution is suitable for real-world deployment. Students should analyze:

- Technical feasibility,
- Infrastructure requirements,
- Computational requirements,
- Data availability,
- Scalability,
- Integration with existing business systems,
- Maintenance considerations,
- Monitoring requirements,
- Model update strategy.

6. **Ethical, Legal, and Risk Assessment**

Students must discuss potential risks associated with deploying the model. Students should evaluate:

- Data privacy considerations,
- Ethical implications,
- Fairness and bias,
- Transparency and explainability,
- Regulatory or compliance considerations,
- Security risks,
- Potential unintended consequences,

- Responsible AI considerations (where applicable).

7. **Final Recommendations and Future Work**

Students must provide evidence-based recommendations based on the project findings. Students should discuss:

- Whether the model is recommended for deployment,
- Situations where deployment may or may not be appropriate,
- Remaining limitations,
- Additional data that could improve performance,
- Possible methodological improvements,
- Future research directions,
- Long-term opportunities for extending the proj

Milestone 6 Deliverables

Students must submit one professionally formatted document (PDF or Word) containing the following sections:

1. Title Page

The title page should include:

- Project Title
- Student Group Name
- Full Names of Team Members

2. Updated Executive Summary (150–250 words)

The updated summary should briefly describe:

- Validation methodology,
- Model robustness,
- Business impact assessment,
- Deployment feasibility,
- Key findings,
- Major limitations,
- Final recommendations.

3. Model Validation

Students must include:

- Validation methodology,
- Validation procedures,
- Test or validation dataset description,
- Generalization results,
- Performance metrics,
- Supporting tables and visualizations.

4. Robustness and Reliability Assessment

Students must document:

- Robustness evaluation,
- Stability analysis,
- Sensitivity analysis,
- Reliability assessment,
- Discussion of validation results.

5. Error Analysis

Students must provide:

- Error distribution,
- Misclassification or residual analysis,
- Error visualizations,
- Root causes of prediction errors,
- Potential improvements.

6. **Business Impact Assessment**

Students should provide:

- Business objectives addressed,
- Operational benefits,
- Financial impact,
- Decision-making improvements,
- Expected organizational value.

7. **Business Interpretation**

Students should explain:

- How project findings support business decisions,
- Expected return on investment (where applicable),
- Practical significance of the model,
- Alignment with project objectives.

8. **Deployment Feasibility Assessment**

Students must discuss:

- Technical deployment requirements,
- Infrastructure considerations,
- Scalability,
- Maintenance requirements,
- Monitoring strategy,
- Practical implementation challenges,
- Deployment recommendations.

9. **Ethical, Legal, and Risk Assessment**

Students must include:

- Ethical considerations,
- Privacy considerations,
- Bias assessment,
- Explainability considerations,
- Regulatory issues (if applicable),
- Risk mitigation strategies.

10. **Final Conclusions and Recommendations**

Students must summarize:

- Overall project achievements,
- Validation outcomes,
- Business impact,
- Deployment readiness,
- Project limitations,
- Recommendations for future work.

11. References (IEEE Style)

Students must use proper IEEE citation formatting consistently throughout the document.

Example References

[1] J. Smith and A. Kumar, "Explainable ensemble learning for customer churn prediction," *IEEE Access*, vol. 13, pp. 11234–11249, 2025.

[2] R. Chen et al., "Comparative analysis of deep learning and traditional machine learning methods for financial forecasting," *Expert Systems with Applications*, vol. 245, 2026.

Submission Guideline

Please submit one **Validation & Business Impact Assessment Report** per group through Canvas no later than **July 21, 2026, at 11:59 PM**.